Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 09)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 1 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 1 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 2 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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2 .....

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0 2 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 2 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 3 The Rio Bravo Conservation and Management Area in Belize is a largely undamaged rainforest with over 200 species of tree, 400 species of bird and 70 species of mammal. It is managed by a Belizean NGO to conserve wildlife, produce resources, sequester carbon and benefit the local economy.

0 3 . 1 Explain why creating the reserve could have threatened local communities, and describe how the project avoids this.

[3 marks]

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0 3 . 2 Explain **four** of the rules followed when trees are selected and felled that keep the exploitation sustainable.

[4 marks]

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0 3 . 3 Each felled tree is given a unique identification tag. Explain how this supports Forest Stewardship Council accreditation.

[2 marks]

0 3 . 4 Describe **three** ways the reserve raises money without damaging the forest.

[3 marks]

0 3 . 5 Explain why selective logging of this kind causes much less ecological damage than clear felling.

[3 marks]

15

0 4 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

Soil maintenance

Interspecies relationships

0 4 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

0 4 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

10

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0 5 Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha^{-1}	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

0 5 . 1 Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

0 5 . 2 Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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0 5 . 3 Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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0 6 . 1 For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food .....

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Fashion .....

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Entertainment / pets .....

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Furniture and ornaments .....

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Traditional medicines .....

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0 6 . 2 Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

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0 6 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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0 7 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 7 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 7 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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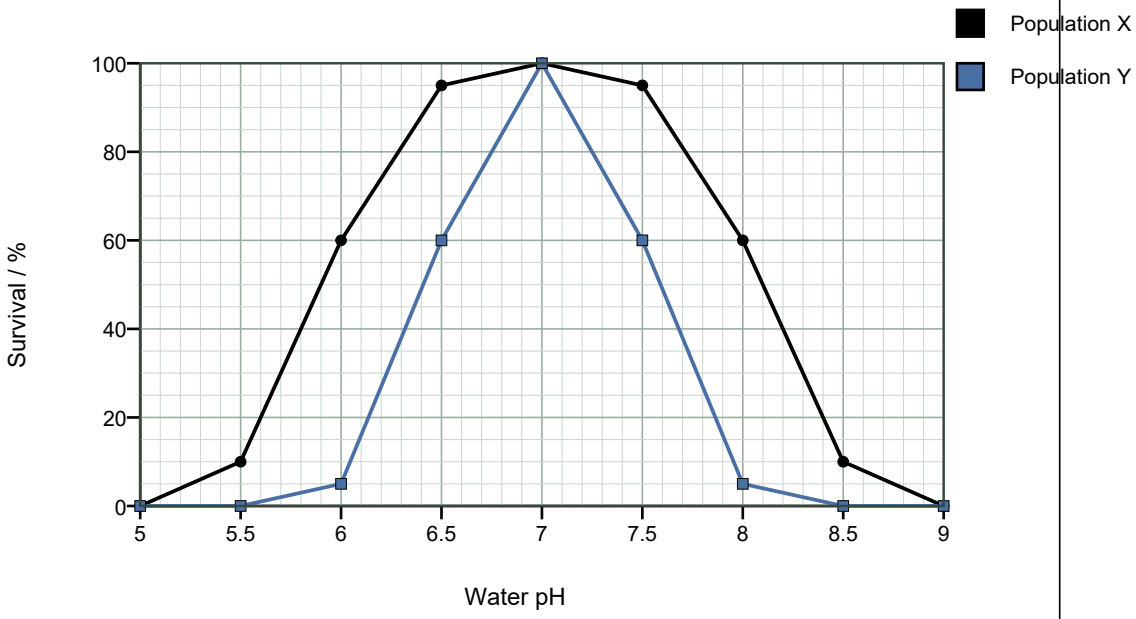
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0 8 Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 8 . 1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

0 8 . 2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

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0 8 . 3 Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

0 8 . 4 Give **two** biotic factors that affect the distribution of a species.

[2 marks]

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0 9 . 1 Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

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0 9 . 2 Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

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0 9 . 3 Describe **three** threats to UK broadleaf woodland.

[3 marks]

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1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ☐ %

1 0 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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1 0 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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1 0 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

1 1 . 2 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS